



DEPARTMENT OF COMPUTER ENGINEERING

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Name and Signature of Faculty with Date

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PROBLEM STATEMENT

The rapid growth of e-commerce platforms, food delivery apps, and online consumer marketplaces has made it very easy for customers to share their dining and grocery experiences. Along with normal positive and negative reviews, customers frequently use sarcasm and figurative irony when expressing dissatisfaction.

Standard machine learning and sentiment analysis models typically rely on counting positive and negative words. These models fail when faced with sarcastic reviews. For example, if a customer writes, "Great job delivering completely sour milk! Loved throwing it in the trash!", a standard bag-of-words algorithm detects words like "Great" and "Loved" and incorrectly classifies the review as Positive.

The aim of this project is to develop a Machine Learning-based Sentiment Analysis and Sarcasm Detection System that can automatically analyze consumer review text, classify it into Positive, Neutral, or Negative categories, and accurately detect sarcasm so that ironic complaints are properly identified and corrected to Negative. The system also provides confidence values and class probabilities along with predictions to help users understand the model's output.

INTRODUCTION

Sentiment analysis refers to the computational identification and categorization of opinions expressed in text to determine the writer's attitude toward a product or service. With the massive growth of e-commerce, customer feedback is generated in huge volumes, making manual review impossible.

Machine Learning can be used to identify patterns in previously labelled reviews and use those patterns to classify new, unseen reviews. In this project, a Food Review Sentiment Analysis and Sarcasm Detection System is developed combining classical statistical machine learning with deep transformer attention.

The system uses the Amazon Fine Food Reviews Dataset for training. The text data is preprocessed and converted into numerical features using TF-IDF (Term Frequency-Inverse Document Frequency) across unigrams, bigrams, and trigrams. A Logistic Regression model is trained to classify baseline customer sentiment into Positive, Neutral, or Negative.

To handle figurative irony, the system integrates a pre-trained RoBERTa transformer model to evaluate the whole sentence context and determine sarcasm probability. If positive words are detected as being used sarcastically, the system corrects the final prediction to Negative.

The trained models are integrated into an interactive web application built using Flask. Users can enter a customer review and receive an instant sentiment prediction along with confidence scores, probability breakdowns, and an explanation of any detected sarcasm.

SPECIFICATION/REQUIREMENTS

Software:

The following software and technologies are used in the project:

Software / Technology	Purpose
Python	Main programming language
Jupyter Notebook / VS Code	Development environment
Pandas	Dataset handling and processing
NumPy	Numerical and array operations
Scikit-learn	Machine Learning implementation
TF-IDF Vectorizer	Text feature extraction (1, 3 n-grams)
Logistic Regression	Sentiment classification
Transformers (RoBERTa)	Deep learning sarcasm detection
Joblib	Saving and loading trained models
Flask	Web application interface

Hardware:

The project can be implemented on a normal computer or laptop without requiring specialized hardware.

- Processor: Intel i3 or above / equivalent
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 2-5 GB free space
- Keyboard and mouse
- Internet connection for installing Python libraries and model weights

ALGPRITHMS/TECHNOLOGY USED

1) Front end:

1. Web User Interface (HTML5, CSS3, JavaScript)

A responsive web dashboard is developed for the frontend of the application. It allows the Python-based Machine Learning models to be presented through a simple and interactive user interface.

The frontend provides the following features:

- Interactive sample buttons to quickly test Positive, Negative, and Sarcastic reviews
- Analyze Review button
- Display of Positive / Neutral / Negative calibrated prediction badge
- Display of model confidence and class probability percentages
- Display of Sarcasm detection status (Literal vs. Sarcastic)
- Visual probability progress bars for both Machine Learning and Deep Learning engines
- Sarcasm Inversion alert box explaining why a decision was changed
-

2)Back end:

1. Python

Python is used as the main backend programming language. It handles dataset processing, text preprocessing, TF-IDF feature extraction, Machine Learning model training, sarcasm inference, and web request routing.

2. Pandas

Pandas is used for loading and processing the Amazon Fine Food Reviews Dataset.

It is used to:

- Load raw review CSV records
- Filter necessary columns (review summary, text, score)
- Drop missing and null entries
- Map 1-5 star ratings into Positive, Neutral, and Negative classes
- Prepare balanced data for Machine Learning

3. NumPy

NumPy is used for numerical operations, array slicing, and handling probability matrices generated during model evaluation.

4. NLP/Text Preprocessing

The backend performs preprocessing on the raw review text before it is given to the Machine Learning model.

The preprocessing includes:

- Converting text to lowercase
- Expanding common contractions (e.g., "didn't" to "did not", "wasn't" to "was not")
- Removing URLs and web links
- Removing HTML markup tags
- Removing punctuation marks and special characters
- Removing extra whitespaces

This produces cleaner text for feature extraction while preserving crucial negation words.

5. TF-IDF Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) is used to convert cleaned review text into numerical feature vectors.

The Machine Learning model cannot directly process raw sentences. Therefore, TF-IDF represents the text numerically based on word importance. It is configured with n-gram range (1, 3) to capture word combinations like "not good" and capped at 75,000 features.

The trained vectorizer is saved as: **tfidf_vectorizer.pkl**

6. Logistic Regression

Logistic Regression is the main Machine Learning algorithm used for sentiment classification.

It is trained using the TF-IDF features and balanced class weights to prevent bias toward the majority class.

The model classifies reviews into three categories:

- Negative (0)
- Neutral (1)
- Positive (2)

The trained model is saved as: **sentiment_model.pkl**

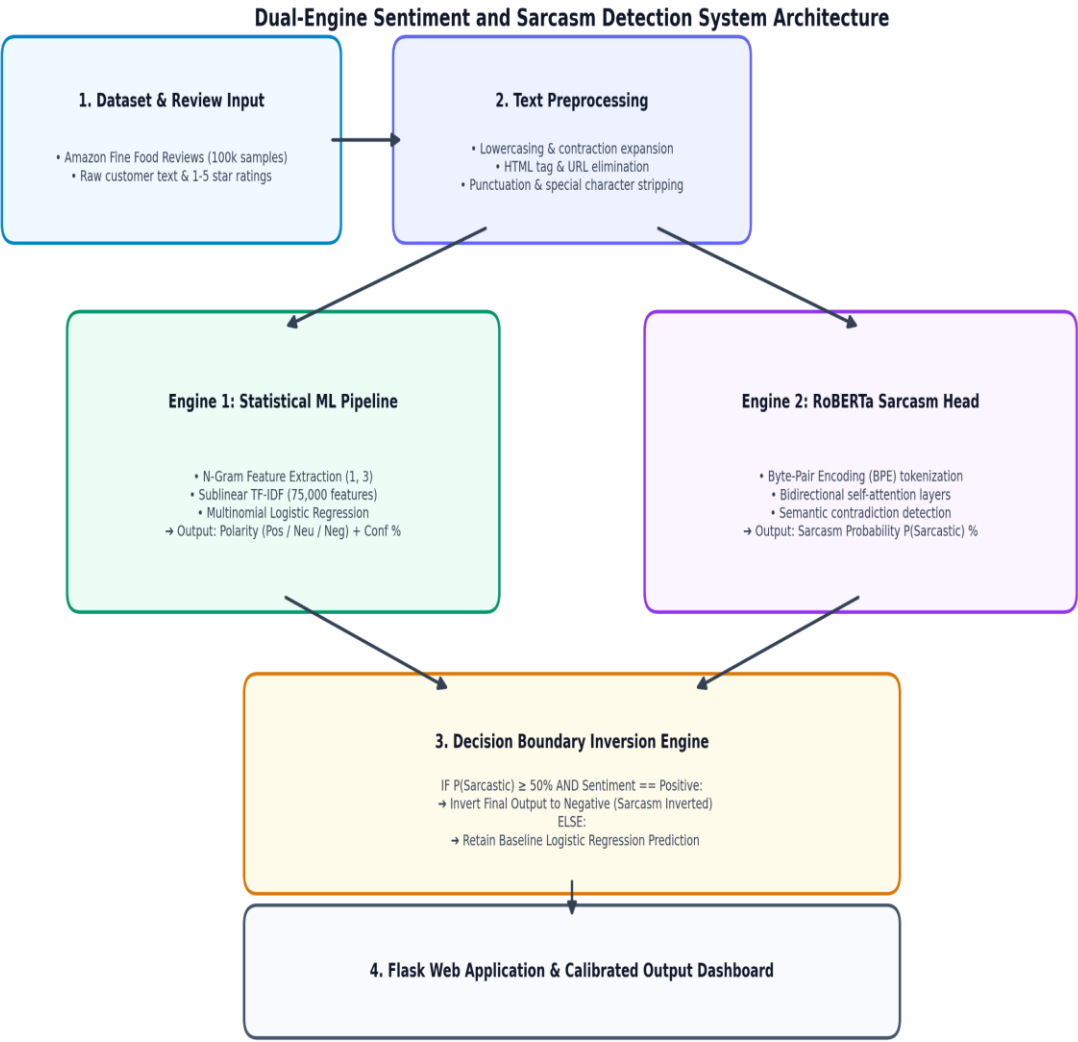
7. RoBERTa Transformer (Sarcasm Detection)

A pre-trained RoBERTa model (twitter-roberta-base-irony) is used to detect figurative sarcasm. It uses bidirectional self-attention to identify semantic contradictions between praise words and product flaws.

8. Scikit-learn

Scikit-learn provides the TF-IDF vectorizer, Logistic Regression classifier, train-test splitting, metrics calculation, and evaluation functionality.

ANALYSIS/DESIGN



IMPLEMENTATION

6.1 Dataset Preparation

The Amazon Fine Food Reviews dataset is loaded using Pandas. Star ratings are converted into sentiment categories: 1-2 stars as Negative (0), 3 stars as Neutral (1), and 4-5 stars as Positive (2). The dataset is shuffled and split into 80,000 training reviews and 20,000 testing reviews.

6.2 Text Preprocessing

The review text is cleaned before model training. Contractions are expanded so that words like "not" are never lost. URLs, HTML tags, punctuation, and extra spaces are removed, and text is converted to lowercase.

6.3 TF-IDF Feature Extraction

The cleaned text content is transformed using the TF-IDF vectorizer with unigrams, bigrams, and trigrams. The vectorizer learns key vocabulary from the training set and converts each review into a numerical vector.

The trained vectorizer is saved as: `tfidf_vectorizer.pkl`

6.4 Model Training

The TF-IDF feature matrices are provided to the Logistic Regression algorithm with balanced class weights. The model learns the statistical relationship between word patterns and sentiment classes.

The trained model is saved as: `sentiment_model.pkl`

6.5 Sarcasm Detection & Inversion Logic

The uncleaned review sentence is also evaluated by the RoBERTa sarcasm model. If the sarcasm confidence is 50% or higher and the baseline ML sentiment is Positive, the system overrides the decision to Negative (Sarcasm Inverted).

6.6 Web Interface

The trained models and decision logic are integrated into a Flask web application. The interface provides a simple way for users to enter reviews and receive instant predictions with confidence metrics.

RESULTS

DUAL-ENGINE INTELLIGENCE

Food Review Sentiment & Sarcasm AI

Evaluates customer sentiment via Logistic Regression (TF-IDF) and catches contextual irony with RoBERTa self-attention.

Try Examples:

Positive

Negative

Sarcastic

10/10 for making me regret my order

Analyze Review

Final Calibrated Sentiment

Negative (Sarcasm Inverted)

Irony Status

Sarcastic (Irony)

ENGINE 1: LOGISTIC REGRESSION

Positive

83.6%

ENGINE 2: ROBERTA SARCASM

Probability

84.3%



Positive vocabulary was used sarcastically to describe an unsatisfactory experience.

